

H524 Introduction to Biostatistics

Week 9: Correlation and Simple Linear Regression

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Fall 2025

Outline

- 1 Review and Introduction
- 2 Correlation (Chapter 17)
- 3 Simple Linear Regression (Chapter 18)
- 4 AI as a Learning Tool for Regression
- 5 Summary and Examples

What We've Learned So Far

Weeks 1-8 covered:

- Descriptive statistics and data visualization
- Probability and diagnostic testing
- Sampling distributions and Central Limit Theorem
- Confidence intervals
- Hypothesis testing (one-sample, two-sample, paired)
- ANOVA (comparing multiple groups)
- Categorical data analysis (chi-square tests)

This Week: Relationships between continuous variables

- How do two variables relate to each other?
- Can we predict one variable from another?

Reading: Pagano & Gauvreau Chapters 17 and 18

New Questions We Can Answer

Previous weeks: Comparisons between groups

- Is mean cholesterol different between treatment groups?
- Is smoking status associated with lung cancer?

This week: Relationships between continuous variables

- How is systolic blood pressure related to age?
- Can we predict birth weight from gestational age?
- Is there a relationship between BMI and cholesterol?
- How strong is the association between height and weight?

Key concepts:

- **Correlation:** Measure strength and direction of linear relationship
- **Regression:** Predict one variable from another

What is Correlation?

Correlation: A measure of the linear association between two continuous variables

Key characteristics:

- Measures strength and direction of relationship
- Ranges from -1 to $+1$
- $r = 0$: No linear relationship
- $r = +1$: Perfect positive linear relationship
- $r = -1$: Perfect negative linear relationship
- $|r| > 0.7$: Strong correlation
- $|r| = 0.3$ to 0.7 : Moderate correlation
- $|r| < 0.3$: Weak correlation

Important: Correlation measures LINEAR relationships only

- Two variables can be strongly related but have $r \approx 0$ if relationship is nonlinear

Scatter plots reveal the relationship:

- **Positive correlation:** As X increases, Y increases
 - Example: Height and weight
 - Points trend upward from left to right
- **Negative correlation:** As X increases, Y decreases
 - Example: Exercise frequency and resting heart rate
 - Points trend downward from left to right
- **No correlation:** No clear pattern
 - Example: Shoe size and IQ
 - Points scattered randomly

Always plot your data first! Summary statistics can mislead without visualization.

Pearson Correlation Coefficient

Most common correlation measure: Pearson's r

Formula:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Alternative formula:

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Interpretation:

- r measures how consistently X and Y vary together
- Standardized measure (unitless)
- Symmetric: $r_{xy} = r_{yx}$

In R: `cor(x, y)` or `cor.test(x, y)`

Example: Age and Systolic Blood Pressure

Research question: Is blood pressure related to age?

Data: 15 adults aged 30-70 years

Subject	Age	SBP	Subject	Age	SBP
1	35	118	9	52	138
2	38	122	10	55	140
3	40	125	11	58	145
4	42	128	12	60	148
5	45	130	13	62	150
6	47	132	14	65	155
7	48	135	15	68	158
8	50	136			

Pearson correlation: $r = 0.98$ (very strong positive)

Interpretation: Age and systolic BP are strongly positively correlated

Testing Correlation Significance

Question: Is the observed correlation statistically significant, or could it be due to chance?

Hypotheses:

- $H_0 : \rho = 0$ (population correlation is zero)
- $H_A : \rho \neq 0$ (population correlation is not zero)

Test statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$$

Example: Age and BP

- $r = 0.98, n = 15$
- $t = \frac{0.98\sqrt{13}}{\sqrt{1-0.98^2}} = \frac{3.535}{0.199} = 17.76$
- $df = 13, p < 0.001$
- **Conclusion:** Very strong evidence of positive correlation

In R: `cor.test(age, sbp)` provides r , t , p -value, and CI

Assumptions for Pearson Correlation

For valid inference, we assume:

- ➊ **Bivariate normality:** (X, Y) follow a bivariate normal distribution
 - Each variable approximately normal
 - Joint distribution is elliptical
- ➋ **Linear relationship:** Association is linear
 - Check with scatter plot
 - Pearson r may miss nonlinear patterns
- ➌ **No extreme outliers:** Outliers strongly influence r
 - Single outlier can create or destroy correlation
 - Always examine scatter plot
- ➍ **Random sample:** Observations independently sampled

If assumptions violated: Consider Spearman's rank correlation (more robust)

Spearman's Rank Correlation

Alternative to Pearson: Based on ranks, not raw values

When to use Spearman's r_s :

- Relationship is monotonic but not linear
- Data contain outliers
- Data are ordinal (ranks)
- Distribution is highly skewed

Method:

- 1 Rank X values: $1, 2, 3, \dots, n$
- 2 Rank Y values: $1, 2, 3, \dots, n$
- 3 Calculate Pearson correlation on ranks

Properties:

- Ranges from -1 to $+1$
- Less sensitive to outliers than Pearson r
- Detects monotonic (not just linear) relationships

CRITICAL WARNING:

Correlation does NOT imply causation!

Examples of spurious correlations:

- Ice cream sales and drowning deaths (both increase in summer)
- Number of fire trucks and damage from fires (both higher for big fires)
- Shoe size and reading ability in children (both increase with age)

Possible explanations for correlation:

- 1 X causes Y
- 2 Y causes X
- 3 Both X and Y caused by third variable (confounding)
- 4 Coincidence

To establish causation: Need randomized controlled experiments, not just observational correlations.

Important Properties of Correlation

What correlation tells us:

- Strength of linear relationship
- Direction (positive or negative)
- Statistical significance

What correlation does NOT tell us:

- Causation (as discussed)
- The form of the relationship (linear? exponential?)
- How to predict Y from X

Key limitations:

- Sensitive to outliers
- Only measures linear relationships
- Same r can arise from very different scatter plots
- Restricted range reduces correlation

Coefficient of determination: r^2

From Correlation to Regression

Correlation: Measures strength of relationship (symmetric)

Regression: Models relationship to make predictions (asymmetric)

Simple linear regression:

- One response variable (Y): outcome we want to predict
- One predictor variable (X): variable we use for prediction
- Assumes linear relationship: $Y = \beta_0 + \beta_1 X + \epsilon$

Goals of regression:

- 1 Describe relationship between X and Y
- 2 Estimate the effect of X on Y
- 3 Predict Y for given values of X
- 4 Test hypotheses about the relationship

Example: Predict birth weight (Y) from gestational age (X)

The Simple Linear Regression Model

Population model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where:

- Y_i : Response variable for subject i
- X_i : Predictor variable for subject i
- β_0 : Population intercept (value of Y when $X = 0$)
- β_1 : Population slope (change in Y per unit change in X)
- ϵ_i : Random error term, $\epsilon_i \sim N(0, \sigma^2)$

Sample regression line (fitted model):

$$\hat{Y}_i = b_0 + b_1 X_i$$

where:

- $\hat{Y}_i$: Predicted value of Y for subject i
- b_0 : Sample estimate of β_0
- b_1 : Sample estimate of β_1

Interpreting Regression Coefficients

Intercept (b_0):

- Predicted value of Y when $X = 0$
- May not be meaningful if $X = 0$ is outside observed range
- Example: Birth weight when gestational age = 0 weeks (nonsensical)

Slope (b_1):

- Change in Y for each 1-unit increase in X
- Most important parameter
- **Example:** $b_1 = 100$ g/week
 - Each additional week of gestation increases birth weight by 100 g
 - 3 weeks longer gestation $\Rightarrow$ 300 g heavier baby (on average)

Key point: Slope quantifies the relationship

- $b_1 > 0$: Positive relationship
- $b_1 < 0$: Negative relationship
- $b_1 = 0$: No linear relationship

Least Squares Estimation

How do we find the best-fitting line?

Method of least squares: Minimize sum of squared residuals

Residual: $e_i = Y_i - \hat{Y}_i$ (observed – predicted)

Objective: Minimize $\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$

Formulas for estimates:

$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = r \times \frac{s_Y}{s_X}$$

$$b_0 = \bar{Y} - b_1 \bar{X}$$

Properties:

- Regression line always passes through $(\bar{X}, \bar{Y})$
- Unique solution (one best-fitting line)
- Sum of residuals equals zero: $\sum e_i = 0$

Example: Age and Blood Pressure Regression

Data: Age (X) and systolic BP (Y) from 15 adults

Summary statistics:

- $\bar{X} = 50$ years, $s_X = 10.49$ years
- $\bar{Y} = 138$ mmHg, $s_Y = 13.42$ mmHg
- $r = 0.98$

Calculate slope:

$$b_1 = r \times \frac{s_Y}{s_X} = 0.98 \times \frac{13.42}{10.49} = 1.25 \text{ mmHg/year}$$

Calculate intercept:

$$b_0 = \bar{Y} - b_1 \bar{X} = 138 - 1.25(50) = 75.5 \text{ mmHg}$$

Regression equation:

$$\widehat{\text{SBP}} = 75.5 + 1.25 \times \text{Age}$$

Interpretation: For each additional year of age, systolic BP increases by

Using the regression equation for prediction:

$$\widehat{SBP} = 75.5 + 1.25 \times \text{Age}$$

Example predictions:

- Age 40: $\widehat{SBP} = 75.5 + 1.25(40) = 125.5$ mmHg
- Age 55: $\widehat{SBP} = 75.5 + 1.25(55) = 144.3$ mmHg
- Age 70: $\widehat{SBP} = 75.5 + 1.25(70) = 163.0$ mmHg

Cautions about prediction:

- **Interpolation:** Predictions within range of observed X (safe)
- **Extrapolation:** Predictions outside range of observed X (risky!)
 - Age 20: $\widehat{SBP} = 75.5 + 1.25(20) = 100.5$ mmHg (OK if reasonable)
 - Age 100: $\widehat{SBP} = 75.5 + 1.25(100) = 200.5$ mmHg (unreliable!)
- Relationship may not hold outside observed range
- Always check if prediction is in plausible range

Residuals and Residual Plots

Residual: Difference between observed and predicted

$$e_i = Y_i - \hat{Y}_i$$

Why residuals matter:

- Measure quality of fit
- Check regression assumptions
- Identify outliers and influential points

Residual plot: Plot e_i vs. $\hat{Y}_i$ (or vs. X_i)

What to look for:

- **Good:** Random scatter around zero, constant spread
- **Problem:** Curved pattern (nonlinearity)
- **Problem:** Funnel shape (non-constant variance)
- **Problem:** Extreme outliers

Always examine residual plots! They reveal model inadequacies.

Assessing Model Fit: R^2

Coefficient of determination: R^2

$$R^2 = \frac{SS_{\text{regression}}}{SS_{\text{total}}} = 1 - \frac{SS_{\text{residual}}}{SS_{\text{total}}}$$

For simple linear regression: $R^2 = r^2$

Interpretation:

- Proportion of variability in Y explained by X
- Ranges from 0 to 1 (or 0% to 100%)
- $R^2 = 0$: X explains none of the variability in Y
- $R^2 = 1$: X explains all of the variability in Y (perfect fit)

Example: Age and BP

- $r = 0.98$, so $R^2 = 0.98^2 = 0.96$ or 96%
- Age explains 96% of the variability in systolic BP
- Remaining 4% due to other factors (genetics, diet, exercise, etc.)

Regression Assumptions

For valid inference, we assume:

- ➊ **Linearity:** True relationship between X and Y is linear
 - Check with scatter plot and residual plot
- ➋ **Independence:** Observations are independent
 - Violated by clustered or repeated measures data
- ➌ **Normality:** Residuals are normally distributed
 - Check with Q-Q plot or histogram of residuals
 - Less critical for large samples (CLT)
- ➍ **Constant variance (homoscedasticity):** Spread of residuals is constant
 - Check with residual plot
 - No funnel shape
- ➎ **No influential outliers:** Extreme points can distort fit
 - Check with residual plot and leverage diagnostics

Inference for Regression Slope

Key question: Is there a significant linear relationship between X and Y?

Hypotheses:

- $H_0 : \beta_1 = 0$ (no linear relationship)
- $H_A : \beta_1 \neq 0$ (linear relationship exists)

Test statistic:

$$t = \frac{b_1 - 0}{SE(b_1)} \sim t_{n-2}$$

where $SE(b_1) = \frac{s}{\sqrt{\sum (X_i - \bar{X})^2}}$ and $s = \sqrt{\frac{\sum e_i^2}{n-2}}$ is the residual standard error

Confidence interval for β_1 :

$$b_1 \pm t_{n-2, \alpha/2} \times SE(b_1)$$

Note: In simple linear regression, testing $\beta_1 = 0$ is equivalent to testing $\rho = 0$

Example: Testing Age-BP Relationship

Regression output from R:

	Estimate	Std. Error	t value	p-value
Intercept	75.5	4.23	17.85	< 0.001
Age	1.25	0.083	15.06	< 0.001

Residual standard error: 2.84 mmHg

$R^2 = 0.946$, Adjusted $R^2 = 0.942$

Interpretation:

- $b_1 = 1.25$ mmHg/year (95% CI: 1.07 to 1.43)
- $t = 15.06$ with $df = 13$, $p < 0.001$
- Very strong evidence that age is associated with systolic BP
- For each year increase in age, SBP increases by 1.25 mmHg (95% CI: 1.07 to 1.43)

In R: `summary(lm(sbp ~ age))`

ANOVA for Regression

Analysis of Variance decomposes total variability:

$$SS_{\text{total}} = SS_{\text{regression}} + SS_{\text{residual}}$$

Source	df	SS	MS	F
Regression	1	1823.7	1823.7	226.8
Residual	13	104.5	8.04	
Total	14	1928.2		

F-test:

$$F = \frac{MS_{\text{regression}}}{MS_{\text{residual}}} = \frac{1823.7}{8.04} = 226.8$$

$p < 0.001$: Strong evidence that the regression model is significant

Note: For simple linear regression, $F = t^2$ and both tests are equivalent.

Prediction Intervals

Two types of intervals for predictions:

1. Confidence interval for mean response:

- Estimates the mean Y for a given X
- Narrower interval
- “What is the average BP for all 55-year-olds?”

2. Prediction interval for individual response:

- Predicts Y for a specific individual at given X
- Wider interval (accounts for individual variation)
- “What BP will this specific 55-year-old have?”

Both intervals:

- Are narrowest at $\bar{X}$
- Widen as we move away from $\bar{X}$
- Should not be used for extrapolation

In R: `predict(model, newdata, interval="confidence")` or

Example: Prediction Intervals

For a 55-year-old:

Point prediction: $\widehat{\text{SBP}} = 75.5 + 1.25(55) = 144.3 \text{ mmHg}$

95% Confidence interval for mean SBP:

- Estimates average SBP for all 55-year-olds
- CI: (142.8, 145.8) mmHg
- Relatively narrow (2-3 mmHg)

95% Prediction interval for individual SBP:

- Predicts SBP for one specific 55-year-old
- PI: (138.0, 150.6) mmHg
- Much wider (12-13 mmHg)
- Accounts for individual variation around the mean

Interpretation: We're 95% confident this individual's SBP will be between 138 and 151 mmHg

Transformations in Regression

When assumptions are violated: Consider transforming variables

Common transformations:

- **Log transformation:** For right-skewed data or multiplicative relationships
 - $\log(Y)$ vs. X : Exponential growth/decay
 - Y vs. $\log(X)$: Logarithmic relationship
 - $\log(Y)$ vs. $\log(X)$: Power relationship
- **Square root:** For count data or moderate skewness
 - $\sqrt{Y}$ vs. X
- **Reciprocal:** For hyperbolic relationships
 - $1/Y$ vs. X

Benefits:

- Linearize relationship
- Stabilize variance
- Improve normality of residuals

Caution: Interpretation changes with transformed variables

Correlation vs. Regression Summary

Correlation	Regression
Symmetric	Asymmetric
Measures strength	Models relationship
No prediction	Makes predictions
r ranges -1 to $+1$	Slope can be any value
Unitless	Has units
$r_{xy} = r_{yx}$	Different X vs. Y roles
Single number	Equation

When to use each:

- **Correlation:** To measure and test strength of association
- **Regression:** To predict, model relationships, estimate effects

Relationship: In simple linear regression, t -test for $\beta_1 = 0$ is equivalent to t -test for $\rho = 0$

R Code: Complete Analysis

Data

```
age <- c(35, 38, 40, 42, 45, 47, 48, 50, 52, 55,  
        58, 60, 62, 65, 68)  
sbp <- c(118, 122, 125, 128, 130, 132, 135, 136, 138,  
        140, 145, 148, 150, 155, 158)
```

1. Correlation

```
cor.test(age, sbp) # r = 0.98, p < 0.001
```

2. Scatter plot

```
plot(age, sbp, main = "Age vs. SBP",  
      xlab = "Age (years)", ylab = "SBP (mmHg)",  
      pch = 19, col = "blue")
```

3. Regression

```
model <- lm(sbp ~ age)  
summary(model) # Detailed output  
abline(model, col = "red", lwd = 2) # Add line to plot
```

4. Diagnostics

```
par(mfrow = c(2, 2))
```

Practical Example: Birth Weight Study

Research question: How does gestational age affect birth weight?

Data: 50 newborns, gestational age 32-42 weeks

Analysis:

- Scatter plot shows strong positive linear relationship
- Pearson correlation: $r = 0.87$ ($p < 0.001$)
- Regression: $\widehat{\text{Weight}} = -2800 + 150 \times \text{Weeks}$
 - Intercept: -2800 g (not meaningful; 0 weeks is impossible)
 - Slope: 150 g/week (95% CI: 130 to 170)
 - $R^2 = 0.76$ (76% of variability explained)

Interpretation:

- Each additional week of gestation increases birth weight by 150 g on average
- Strong evidence of positive linear relationship ($p < 0.001$)
- Can predict birth weight from gestational age with reasonable accuracy

Practical Example: Exercise and Heart Rate

Research question: Is exercise frequency associated with resting heart rate?

Data: 30 adults, exercise sessions per week (0-7), resting HR (bpm)

Analysis:

- Scatter plot shows negative linear trend
- Pearson correlation: $r = -0.68$ ($p < 0.001$)
- Regression: $\widehat{HR} = 78 - 2.5 \times \text{Sessions}$
 - Intercept: 78 bpm (resting HR with no exercise)
 - Slope: -2.5 bpm/session (95% CI: -3.8 to -1.2)
 - $R^2 = 0.46$ (46% explained)

Interpretation:

- Each additional exercise session per week reduces resting HR by 2.5 bpm
- Moderate negative association (still significant)
- Other factors also influence resting HR (54% of variance unexplained)

Common Pitfalls in Regression

Mistakes to avoid:

1 Ignoring assumptions

- Always check residual plots
- Don't force linear model on nonlinear data

2 Extrapolation

- Don't predict outside range of observed X
- Relationship may not hold in unobserved region

3 Causation claims

- Regression shows association, not causation
- Need experiments for causal inference

4 Ignoring outliers

- Single outlier can drastically change regression line
- Investigate influential points

5 Over-interpreting R^2

- Low R^2 doesn't mean relationship is unimportant
- Clinical significance $\neq$ statistical significance

Summary

Today we covered:

- **Chapter 17: Correlation**
 - Pearson and Spearman correlation
 - Testing significance
 - Correlation vs. causation
- **Chapter 18: Simple Linear Regression**
 - Least squares estimation
 - Interpreting slope and intercept
 - Inference for regression coefficients
 - Prediction and prediction intervals
 - Checking assumptions with residual plots

Key formulas:

- Regression line: $\hat{Y} = b_0 + b_1X$
- Slope: $b_1 = r \times s_Y/s_X$
- $R^2 = r^2$ (proportion of variance explained)

Next Week: Multiple regression (more than one predictor)

Key Takeaways

- 1 **Always visualize first:** Scatter plots reveal patterns
- 2 **Correlation measures association:** But not causation
- 3 **Regression quantifies relationships:** Allows prediction
- 4 **Check assumptions:** Use residual plots
- 5 **Interpret slope in context:** What does 1-unit change in X mean?
- 6 **R^2 shows explained variance:** But low R^2 can still be important
- 7 **Don't extrapolate:** Stay within range of data
- 8 **Understand uncertainty:** Report confidence/prediction intervals

In R:

- `cor.test(x, y)` for correlation
- `lm(y ~ x)` for regression
- `plot(model)` for diagnostics
- `predict(model, newdata)` for predictions

Thank you!

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